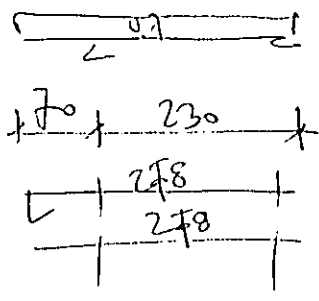


חשבון יציבות: לסימון - תיקון בגודל כח 58

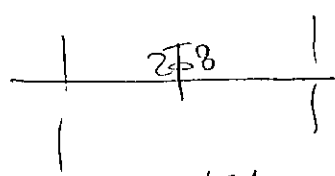
$\sqrt{3}$



$$M = 0.7 \times 0.7^2 / 2 = 0.17 \text{ t-m}$$

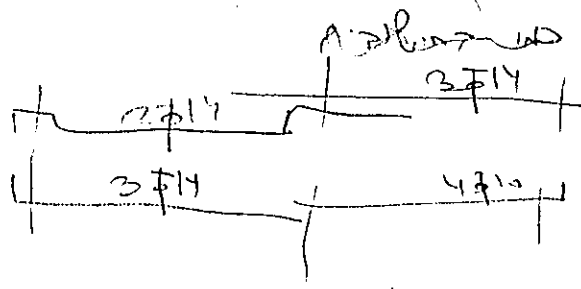
$$A_s^{(+)} = 0.65$$

$\sqrt{3}$



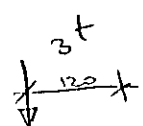
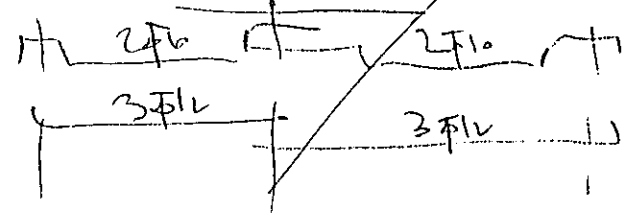
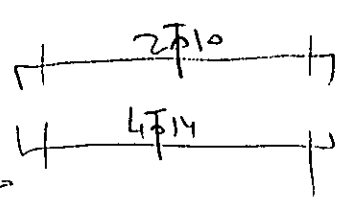
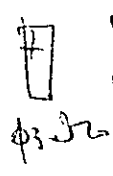
$$A_s = 0.8$$

קורה 5.6-5.7

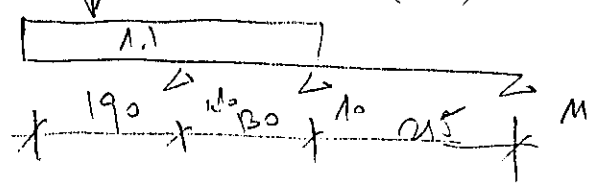


20/80

קורה 8-7-10

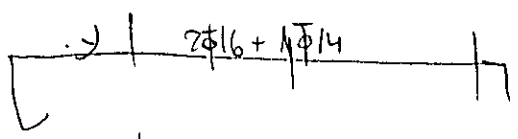


קורה 10-10-10

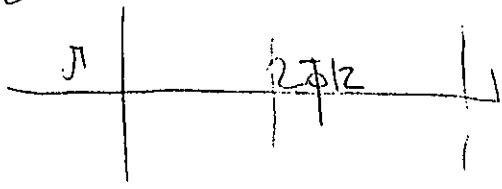


$$M_{1/2} = 3 \times 1.2 + 11 \times 1.9^2 / 2 = 3.6 + 2.56 \text{ t-m}$$

$$A_s = 4.8$$



$$R = 3 + 1.1 \times 1.9 + 5.6 / 13 + 11 \times 0.65 =$$



$$R = 3 + 2.1 + 4.3 + 0.8 = 10.1$$

אוצרת נמוכה

יעקב לבני מהנדס בנין	ערוך ע"י 7/1	תאריך: 18.2.86	דף מס': 2
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חשבון יציבות: קצת - תנאי קליט א.

$A_{10-5} = 5.93$ כ"ס
 $A_{5-6} = 6.82$ "

$T = \frac{4.710}{0.8 \times 16} = 6.9$ מ"מ

קצה 10-5.6

$$0.5 \times 3 = 0.45$$
$$0.67 - 1.5 = 1.0$$
$$\frac{0.3}{1.8}$$

קצה 5-6

$$0.7 \times 1.4 = 0.98$$
$$0.7 \times 1.1 = 0.77$$
$$2.5 \times 1.5 = 3.75$$

$T = \frac{4.100}{0.8 \times 16 \times 5} = 5.7$ מ"מ

$A_{10-7} = 4.26$ כ"ס
 $A_{7-8} = 6.82$ "
 $A_{5-7-8} = 3.38$ "

קצה 5-7-8 (5/20)

$$0.7 \times 1.4 = 0.98$$
$$0.7 \times 1.1 = 0.77$$
$$2.5 \times 1.5 = 3.75$$

$T = \frac{4.100}{0.8 \times 16 \times 5} = 5.7$ מ"מ

$M = 0.8 \times 3^2 / 8 = 0.9$ ת"מ
 $A_5 = 2.64$ כ"ס

קצה 3-20

$$0.4$$
$$0.6$$
$$0.5 \times 1.5 = 0.75$$
$$1.2$$

$M = 2.6 \times 2.6 + 1.2 \times 2.6^2 / 2 = 7.4 + 4.05 = 11.45$ ת"מ
 $A_5 = 7.0$ כ"ס

קצה 20/80 - 10-M

$$0.4$$
$$0.6$$
$$0.5 \times 1.5 = 0.75$$
$$1.2$$

$R_M^{(-)} = 1.2 \times 1.05 - 11.45 / 2 = -4.4$ ת

יעקב לבני מהנדס בנין	עורך ע"י [Signature]	תאריך: 19/11/86	דף מס': 2
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חשבון יציבות:

לפני: יקרה חסר +288

קורה 10-10 (20/90) קורה איינה

$M = 1.0 \times 1.3 + 1.2 \times 1.3^2 =$
 $M = 1.3 + 1.14 = 2.44$

$A_s = 7.74$

$\tau = \frac{M}{S} = \frac{2.44}{0.08} = 30.5$
 ק"מ/מ"מ

לפי: 3.0
 קורה ניצב

$R_m = 1.2 \times 2.0/2 - 1.4/2.1 = -5.4$
 $+ 2.36$
 $\underline{\underline{3.0}}$ t

קורה ניצב

לפי: 3.0 קורה ניצב
 לפי: 3.0 קורה ניצב

חשבון יציבות:

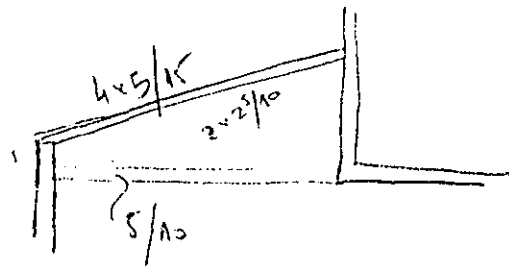
לנצח

קוואר הגג

קוואר הגג

$$M = 0.18 \times 31^2 / 8 = 0.18 \text{ tm} \quad W_{plz} = 0.18 \times 1.5 / 10 = 180 \text{ cm}^3$$

$$W_{plz} = 0.18 \times 1.5 / 10 = 180 \text{ cm}^3 \quad W_{plz} = 0.18 \times 1.5 / 10 = 180 \text{ cm}^3$$



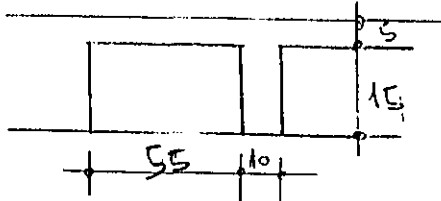
א. 2.10

$$M = 0.18 \times 2.1^2 / 8 = 0.18 \text{ tm} \quad W_{plz} = 17.7 \text{ cm}^3$$

קוואר

2.10

55/30/15 2/10/10 1.10/1.10 d = 14 + 5 = 19



$$2400 \left(0.05 + \frac{0.1 \times 0.15}{0.65} \right) = 176 \text{ cm}^3$$

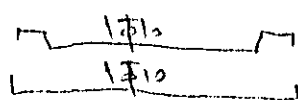
$$500 \times 0.65 \times 0.15 = 63 \text{ cm}^3$$

$$260 \text{ cm}^3$$

$$150 \text{ cm}^3$$

א. 2.10

$$M = 0.18 \times 3^2 / 8 = 0.18 \text{ tm} \quad A_s = 1.39 \text{ cm}^2$$

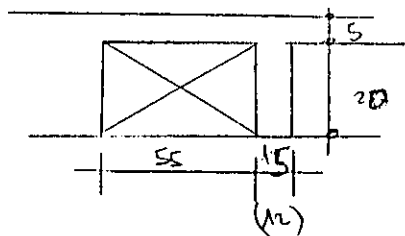


$$0.65 \times \frac{0.65}{0.67} = 1.05$$

חשבון יציבות:

$$d = 20 + 5 = 25$$

תקרה 13 וילוי גילוי סליל



$$2400 \left(0.05 + \frac{0.15 \times 0.2}{0.7} + \frac{0.15 \times 0.2}{2.5} \right) = 252$$

$$500 \times 0.55 \times 0.2 = 79$$

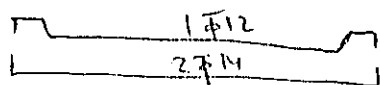
$$260$$

$$150$$

$$740$$

$$M = 1.38 \times 15.55 \times \frac{2}{3} = 285 \text{ tm} \quad A_s = 4.27$$

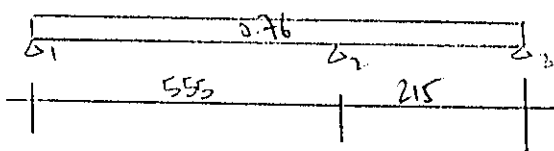
$$\sqrt{13}$$



$$T = \frac{2100 \times 0.65}{0.5 \times 15 \times 22} = 4.6$$

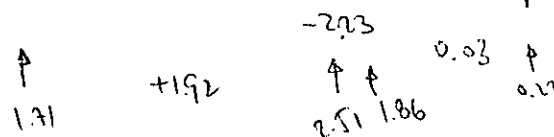
$$b = 12$$

$$\sqrt{13}$$

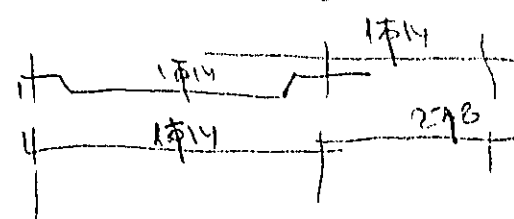


$$A_{s1} = 2.54 \quad \sqrt{2.67}$$

$$A_{s2} = 2.95 \quad \sqrt{2.1}$$



$$T = \frac{2510 \times 0.62}{0.8 \times 12 \times 22} = 6.55$$

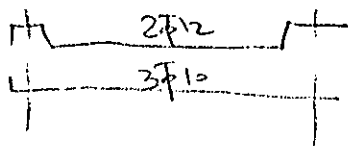


$$\frac{0.74}{0.76} \times \frac{0.67}{0.62} = 1.05$$

$$M = 1.38 \times 3.5 \times \frac{2}{3} = 2.07 \text{ tm}$$

$$(4.25) \quad 11-12 \quad \sqrt{13}$$

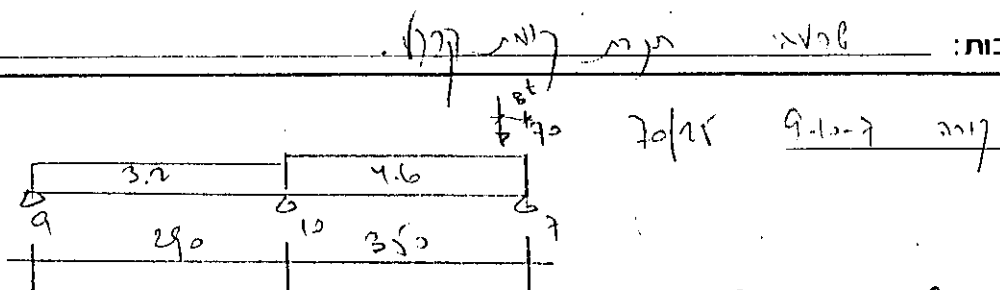
$$A_s = 4.42$$



$$R = 2.36$$

אצטות נאצרת

חשבון יציבות:



$$\begin{array}{r} 0.55 \\ -3.36 \\ -3.50 \\ \hline 6.86 \text{ tm} \end{array}$$

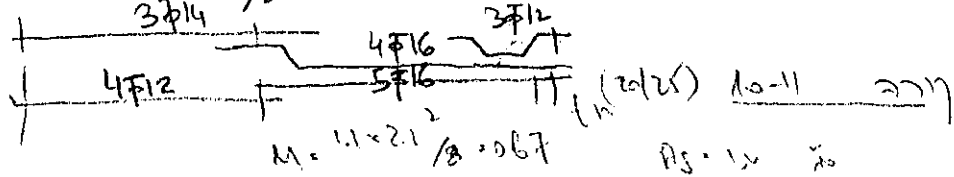
$$M_{10-8} = 4.6 \times 3.5^2 / 8 + \frac{8 \times 0.7 \times 7.8}{2 \times 3.5^2} (0.7 + 3.1) = 7.04 + 2.70 = 9.73$$

$$V_7 = \frac{8 \times 2.8}{3.5} + 4.6 \times 1.75 - \frac{6.86}{3.5} = 12.5 \text{ t}$$

$$V_9 = 3.2 \times 1.75 - \frac{6.86}{2.90} = 4.64 - 2.37 = 2.27 \text{ t}$$

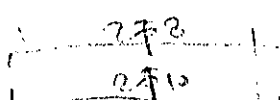
$$M_{9-10} = \frac{2.27^2}{6.4} = 0.8 \text{ tm}$$

$$M_{10-7} = 0.97 \times 12.5 - \frac{4.6 \times 0.97^2}{2} - 0.7 \times 8.0 = 12.22 - 2.16 - 2.16 = 7.9 \text{ tm}$$



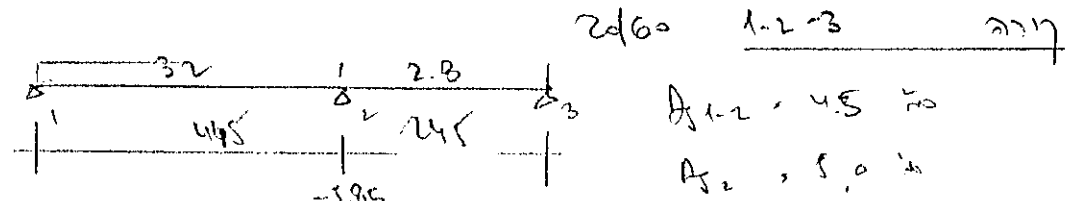
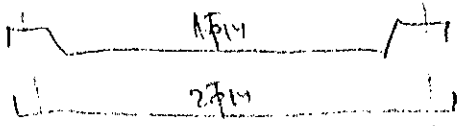
$$M = 1.1 \times 2.1^2 / 8 = 0.67$$

$$A_5 = 1.1 \text{ m}$$



$$M = 1.1 \times 5.5^2 / 8 = 4.16 \text{ tm}$$

$$A_2 = 4.35 \text{ m}$$

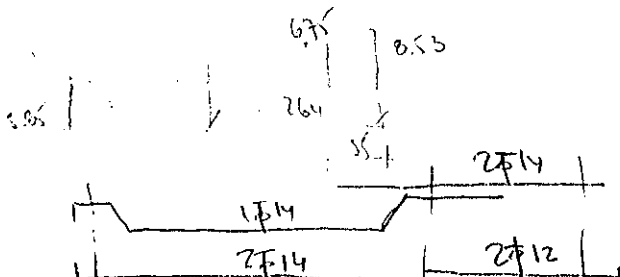


$$A_{1-2} = 4.5 \text{ m}$$

$$A_2 = 5.0 \text{ m}$$

$$T_1 = \frac{844}{1.09 \times 20 \times 55} = 8.53 \text{ t}$$

$$T_2 = \frac{580}{0.9 \times 10 \times 55} = 1.85 \text{ t}$$



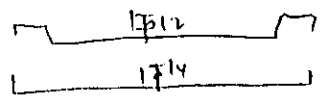
אצות נמורה

תאריך: 9/1-36	ערך ע"י: 7/1	שקב לבני מהנדס בנין
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דף מס' 6

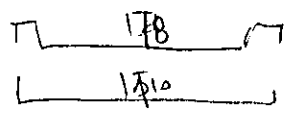
חשבון יציבות: סכום - כבס - חלל - קצוות

כבס 15 ט"ר $\phi = 15 + 5 = 20$ חלל 15 ט"ר $q = 3.15$
 $F_k = 0.65 \text{ t/m}^2$
 $M = 0.65 \times 4.2^2 / 8 = 1.43 \text{ tm}$ $A_s = 2.58$ $\sqrt{3}$



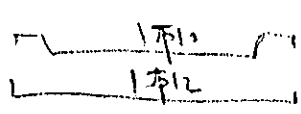
$\sqrt{3}$

$M = 0.65 \times 3^2 / 8 = 0.75 \text{ tm}$ $A_s = 1.31$

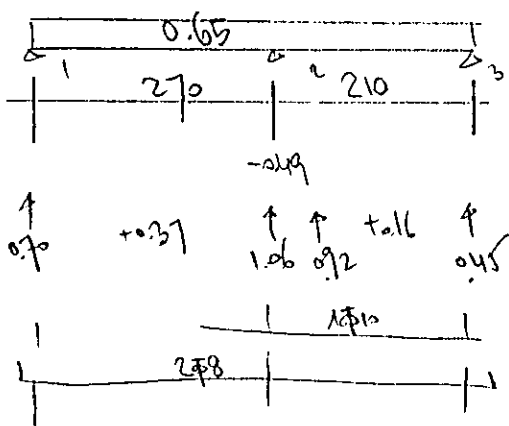


$\sqrt{3}$

$M = 0.65 \times 3.6^2 / 8 = 1.05 \text{ tm}$ $A_s = 1.89$

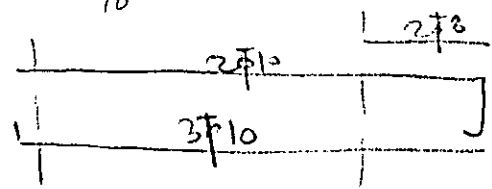


$\sqrt{3}$



$A_{s2} = 0.66$

חבס 1-2 $(2/5, 2/5)$
 $M = 1 \times 4.1^2 / 8 = 2.31 \text{ tm}$ $A_s = 2.41$

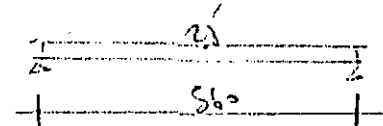
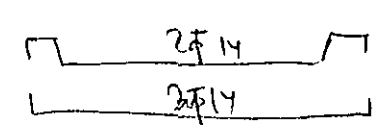


ייעקב לבני מהנדס בנין	ערוך ע"י י. ש.	תאריך: 8/1/9	דף מס': 7
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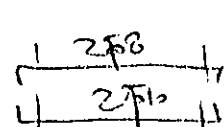
חשבון יציבות: ב-1 כ-2 ג-3 ד-4

קורה 1-9 25/7

$M = 7.5 \times 4.2^2 / 8 = 9.8 \text{ tm}$ $A_s = 7.1$

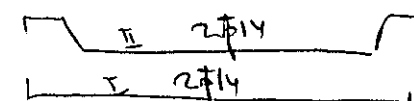



קורה 10-11 20/65



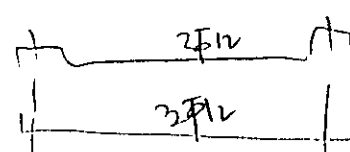
קורה 12-13 20/65

$M = 3.2 \times 4.2^2 / 8 = 7.06 \text{ tm}$ $A_s = 5.6$



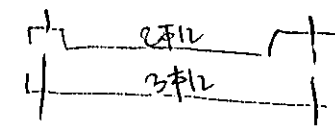
קורה 14-15 40/20

$M = 11 \times 3.6^2 / 8 = 1.78 \text{ tm}$ $A_s = 4.93$

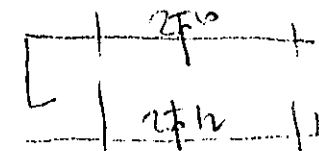


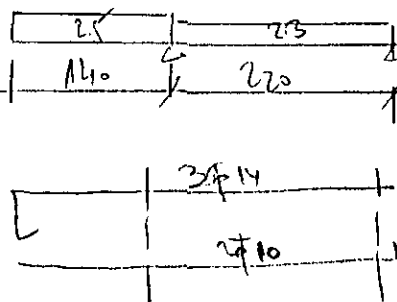
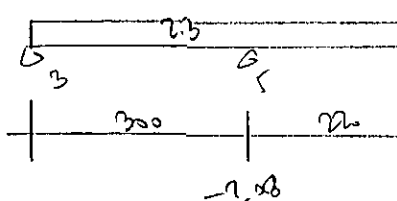
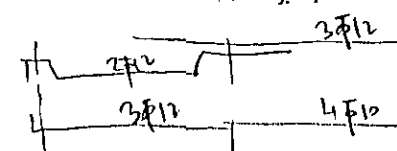
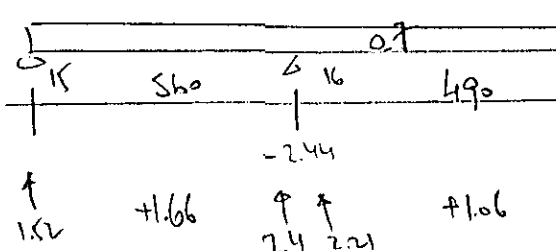
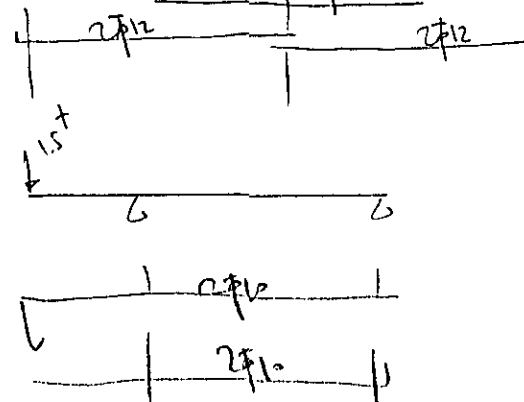
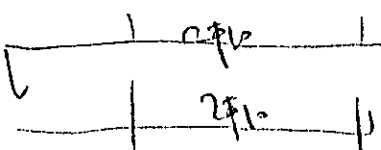
קורה 16-17 40/20

$M = 1.6 \times 3^2 / 8 = 1.8 \text{ tm}$



קורה 18-19 20/65



ייעקב לבני מהנדס בנין	ערור ע"י י.פ.ק.	תאריך: 9/10/86	דף מס' 8
חשבון יציבות: שוואי - צפון-הדרום			
	חלק 11-10 - 20/30 $M^- = 2.45 \times 2.5 = 6.125 \text{ tm}$ $A_s = 3.8$ $M^+ = 3.5 \times 3 = 10.5 \text{ tm}$		
	חלק 7-5-3 - 20/40 $A_{s, 3-5} = 4.56$ $A_{s, 5} = 5.76$		
	חלק 17-6 - $\frac{4.1 \times 1.7}{1.7 \times 1.7} = 1.7$		
	חלק 17-16-15 - (20/40) $A_{s, 15-16} = 2.23$ $A_{s, 16} = 3.28$		
	חלק 1-5 - 20/40 $M = 1.5 \times 0.7 = 1.05 \text{ tm}$ $A_s = 1.4$		
	חלק 9-16 - (20/40) $M^- = 4.6 \times 0.7 = 3.22 \text{ tm}$ $A_s = 4.2$		

דף מס' 9	תאריך: 19.2.86	ערוך ע"י: אב	שקב לבני מהנדס בנן
חשבון יציבות: גרמי - 11.11.86			
①	ג'ג תקיל ג'ג	2.6 17.7 $\frac{1.0}{21.3}t$	$2.5 \times 2.3 + 2.8 = 2.6$ $2 \times (5.3 + 1.1 \times 2.3) = 17.7$ 1.0
②	ג'ג תקיל ג'ג	2.1 78.6 $\frac{1.0}{21.7}t$	$0.5 \times 2 \times 2.6 = 2.1$ $2(8.5 + 5.8) = 78.6$ 1.0
③	ג'ג תקיל ג'ג	4.75 5.5 $\frac{1.0}{13.7}t$	$0.5 \times (2.3 + 1.2) = 4.75$ $2 \times 1.1 + 2 \times 1.5 \times 1.2 = 5.5$ $2 \times 1.1 \times 1.2 + 1.5 \times 1.2 = 5.5$
④	ג'ג תקיל ג'ג	3.2 5.5 1.2	$2 \times 1.1 \times 1.2 = 3.2$ $2 \times 1.1 \times 2.3 = 5.5$ $0.7 \times 1.5 \times 1.2 = 1.2$
⑤	ג'ג תקיל ג'ג	13 12.2	$0.5 \times 2.5 = 13$
⑥	ג'ג תקיל ג'ג	2.2 11.9 $\frac{1.0}{19.3}t$	$0.3 \times 1.1 \times 1.4 \times 2.3 + 1.4 = 2.2$ $2 \times (1.9 \times 5.2 \times 0.6) = 11.9$ 1.0
⑦	ג'ג תקיל ג'ג	2.2 10.9 $\frac{1.0}{13}t$	$0.3 \times 1.1 \times 1.4 \times 2.3 + 1.4 = 2.2$ $2 \times (1.9 \times 5.2 \times 0.6) = 10.9$ 1.0
⑧	ג'ג תקיל ג'ג	10.8 10.8 $\frac{1.0}{21.6}t$	$2.7 \times 2 + 2.7 \times 2 = 10.8$ 10.8 $2.7 \times 2 + 2.7 \times 2 = 10.8$
⑨	ג'ג תקיל ג'ג	2.2 11.4 $\frac{1.0}{15.6}t$	$0.5 \times 4.4 = 2.2$ $2 \times (2.8 + 1.1 \times 2.3) = 11.4$ 1.0
⑩	ג'ג תקיל ג'ג	19.1 34.8 $\frac{1.2}{55.1}t$	$8.3 + 2.7 + 1.2 \times 7.35 + \frac{11}{2.1} = 19.1$ $2 \times (6.5 + 9.6 + 1.1 \times 1.2) = 34.8$ 1.2
⑪	ג'ג תקיל ג'ג	7.4 1.3 $\frac{1.0}{1.3}t$	$(1.1 \times 1.2 + 1.35 \times 1.75) \times 2 = 7.4$ 1.3 1.3
⑫	ג'ג תקיל ג'ג	0.3 4.7 $\frac{1.0}{4.7}t$	$0.3 \times 1.1 \times 1.4 \times 2.3 + 1.4 = 0.3$ 4.7 4.7
⑬	ג'ג תקיל ג'ג	1.8 10.9 $\frac{1.0}{19.3}t$	$0.5 \times 6 \times 0.6 = 1.8$ 10.9 10.9
⑭	ג'ג תקיל ג'ג	7.6 20.3 $\frac{1.0}{20.3}t$	$2 \times (3.5 + 1.75 + 1.1 \times 1.3) = 7.6$ 20.3 20.3



יעקב לבני מהנדס בנין	ערוך ע"י י.ב.	תאריך: 19.2.86	דף מס' 10
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חשבון יציבות: שטח אגמי באן סלול

(13) $\frac{10.9}{16.4} = 0.66$
 $2 \times 11.1 \times 2.5 = 55$
 $\frac{55}{16.4} = 3.35$
2.6

עיני ביטוח 2 גמ. גמולת גמ 3 גמולת, חסר, 2.5 גמולת

$P_{400} \cdot 1.800 = 22.6^t$
 $P_{450} \cdot 1.800 = 29.7^t$
 $P_{460} \cdot 1.800 = 30.8^t$

$P_{470} \cdot 1.800 = \frac{0.7^2 \times 3.14 \times 20}{4} + 2 \times 2.7 \times 3.14 \times 6 = 11.5 + 33 = 44.5^t$
 $P_{480} \cdot 1.800 = \frac{0.8^2 \times 3.14 \times 20}{4} + 2 \times 2.8 \times 3.14 \times 6 = 15.1 + 37.7 = 52.8^t$

קוטר	ל. גמולת	ל. גמולת	באן סלול
430	1.600	4 פ"מ	14
440	1.800	5 פ"מ	1, 3, 4, 6, 9, 11, 14
450	1.800	6 פ"מ	5, 7, 8, 12, 13
460	1.800	8 פ"מ	2,
470	1.800	10 פ"מ	10
480	1.800	11 פ"מ	10

ארכיון נמורה